[}20240769_ _hw1

2026-10-13

```
# Problem 1

## (a) Data Generation
```

```
set.seed(1)

n = 200
x = seq(0, 1, length.out = n)
y = sin(2*pi*x) + rnorm(n, sd = 0.15)

df = data.frame(x, y)

head(df)
```

```
##             x           y
## 1 0.000000000 -0.09396807
## 2 0.005025126  0.05911505
## 3 0.010050251 -0.06223866
## 4 0.015075377  0.33387193
## 5 0.020100503  0.17538587
## 6 0.025125628  0.03414379
```

‘‘‘